

Engineering Mathematics: Matrix Theory

Cayley-Hamilton Theorem — Part 1: Conceptual Foundation

The Cayley-Hamilton Theorem is a fundamental bridge between linear algebra and polynomial algebra. It allows us to treat a matrix as a variable within its own characteristic equation.

1. The Core Statement

Theorem: Every square matrix satisfies its own characteristic equation.

If we have a square matrix A of order n , and its characteristic polynomial is defined by:

$$P(\lambda) = |A - \lambda I| = a_0\lambda^n + a_1\lambda^{n-1} + a_2\lambda^{n-2} + \dots + a_n$$

The theorem states that if you replace the scalar λ with the matrix A (and the constant term a_n with $a_n I$), the result is the Null Matrix (O):

$$a_0 A^n + a_1 A^{n-1} + a_2 A^{n-2} + \dots + a_n I = O$$

2. Conceptual Breakdown

To understand why this is non-trivial, we must look at the components:

1. Characteristic Matrix ($A - \lambda I$):

- This is formed by subtracting λ from the diagonal elements of matrix A .
- Example for a 2×2 matrix:

$$\begin{bmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{bmatrix}$$

2. Characteristic Polynomial:

- When you find the determinant $|A - \lambda I|$, you get a polynomial in terms of λ .
- The degree of this polynomial is equal to the order of the matrix (n).

3. The "Satisfies" Logic:

- Usually, we solve $|A - \lambda I| = 0$ to find Eigenvalues (scalars).
- Cayley-Hamilton tells us that the matrix A itself "plugs into" that formula and results in zero.

3. Step-by-Step Algorithm (Verification)

To verify this theorem for a given matrix A :

- Find the Characteristic Equation:** Calculate the determinant $|A - \lambda I| = 0$.
- Expand the Determinant:** Write it in the form $a_0\lambda^n + a_1\lambda^{n-1} + \dots + a_n = 0$.
- Substitute Matrix A :** Replace λ with A . Note: a_n becomes $a_n I$.
- Compute Powers:** Calculate $A^2, A^3, \dots, A^n$ as required.
- Simplify:** Sum the matrices to verify they result in the Null Matrix.

4. Illustrative Example (2×2)

Let us verify the theorem for matrix

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

Step 1: Characteristic Equation

$$|A - \lambda I| = \begin{vmatrix} 1 - \lambda & 2 \\ 3 & 4 - \lambda \end{vmatrix} = 0$$

$$(1 - \lambda)(4 - \lambda) - (2)(3) = 0$$

$$4 - \lambda - 4\lambda + \lambda^2 - 6 = 0$$

$$\lambda^2 - 5\lambda - 2 = 0$$

Step 2: Cayley-Hamilton Substitution

According to the theorem, A must satisfy:

$$A^2 - 5A - 2I = O$$

Step 3: Verification

Calculate A^2 :

$$A^2 = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \longrightarrow \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix}$$

Substitute into the equation:

$$\begin{aligned} & \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix} - 5 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - 2 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ \longrightarrow & \begin{bmatrix} 7 - 5 - 2 & 10 - 10 - 0 \\ 15 - 15 - 0 & 22 - 20 - 2 \end{bmatrix} \longrightarrow \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \end{aligned}$$

Result: The theorem is verified.

Cayley-Hamilton Theorem — Part 2: The Formal Proof

The proof of this theorem relies on the relationship between a matrix, its adjoint, and its determinant. We use the property: $D \cdot \text{adj}(D) = |D| \cdot I$.

1. The Setup

Let A be a square matrix of order n . Its characteristic equation is:

$$|A - \lambda I| = a_0 \lambda^n + a_1 \lambda^{n-1} + \dots + a_n = 0$$

Now, consider the matrix $B = \text{adj}(A - \lambda I)$. Since the elements of $(A - \lambda I)$ are at most of first degree in λ , the cofactors (and thus the elements of the adjoint) will be polynomials in λ of degree at most $(n - 1)$.

Therefore, we can write the adjoint as a matrix polynomial:

$$\text{adj}(A - \lambda I) = B_0 \lambda^{n-1} + B_1 \lambda^{n-2} + \dots + B_{n-1}$$

Where $B_0, B_1, \dots$ are square matrices of order n .

2. The Derivation

Using the fundamental property $D \cdot \text{adj}(D) = |D| \cdot I$, and substituting $D = (A - \lambda I)$:

$$(A - \lambda I) \cdot \text{adj}(A - \lambda I) = |A - \lambda I| \cdot I$$

Substitute the polynomial forms into this equation:

$$(A - \lambda I)(B_0\lambda^{n-1} + B_1\lambda^{n-2} + \dots + B_{n-1}) = (a_0\lambda^n + a_1\lambda^{n-1} + \dots + a_n)I$$

Equating coefficients of like powers of λ on both sides:

- $\lambda^n : -IB_0 = a_0I$
- $\lambda^{n-1} : AB_0 - IB_1 = a_1I$
- $\lambda^{n-2} : AB_1 - IB_2 = a_2I$
- $\dots$
- $\lambda^0 : AB_{n-1} = a_nI$

3. Final Step: Pre-multiplication

To eliminate the B matrices, we pre-multiply the equations by $A^n, A^{n-1}, A^{n-2}, \dots, I$ respectively:

- $-A^n B_0 = a_0 A^n$
- $A^n B_0 - A^{n-1} B_1 = a_1 A^{n-1}$
- $A^{n-1} B_1 - A^{n-2} B_2 = a_2 A^{n-2}$
- $\dots$
- $AB_{n-1} = a_n I$

Adding all these equations, the terms on the left side cancel out (telescoping sum), leaving:

$$O = a_0 A^n + a_1 A^{n-1} + a_2 A^{n-2} + \dots + a_n I$$

Proof Complete.

Cayley-Hamilton Theorem — Part 3: Engineering Applications

In Engineering M1, the theorem is a powerful tool for rapid matrix computations.

1. Finding the Inverse of a Matrix (A^{-1})

If the constant term $a_n \neq 0$, the matrix is non-singular. We can find the inverse without the full adjoint method.

The Algorithm:

1. Start with the CH Equation: $a_0 A^n + a_1 A^{n-1} + \dots + a_n I = O$.
2. Multiply the entire equation by A^{-1} :

$$a_0 A^{n-1} + a_1 A^{n-2} + \dots + a_{n-1} I + a_n A^{-1} = O$$

3. Rearrange to solve for A^{-1} :

$$A^{-1} = -\frac{1}{a_n} (a_0 A^{n-1} + a_1 A^{n-2} + \dots + a_{n-1} I)$$

2. Finding Higher Powers of Matrices

CH Theorem allows you to reduce high powers (like A^{10}) into a linear combination of $I, A, A^2, \dots, A^{n-1}$.

The Logic:

- Example: If $A^2 - 5A - 2I = O$, then $A^2 = 5A + 2I$.
- To find A^3 , multiply by A : $A^3 = 5A^2 + 2A$.
- Substitute A^2 back in: $A^3 = 5(5A + 2I) + 2A = 27A + 10I$.

3. Summary for Exams

Task	Method
Verify CH Theorem	Find Char. Eq., plug in A , show it equals O .
Find A^{-1}	Multiply CH Eq. by A^{-1} and isolate A^{-1} .
Find A^k	Express A^k as a polynomial of lower powers using the CH Eq.